Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have not specified the topic you want to write about. Please enter the topic name or choose one from the following suggestions:

- History of Artificial Intelligence
- Basics of Quantum Computing
- Introduction to Blockchain Technology
- Principles of Economics
- Fundamentals of Music Theory

Workshop Practice Lab:

- Workshop practice lab is a common theory/lab course for all engineering students in the first and second semester .
- The objective of this lab is to get a hands-on knowledge of several workshop practices like carpentry, fitting, welding, machining, etc. and learn safety regulations to be maintained in a shop floor.
- Workshop practice is the backbone of the real industrial environment which helps to develop and enhance relevant technical hand skills required by the technician working in the various engineering industries and workshops.
- The lab syllabus may vary depending on the university or college, but some of the common experiments are :
  - Introduction to workshop tools and safety precautions
  - Carpentry: making various joints and patterns using wood and nails
  - Fitting: filing, sawing, drilling, tapping and making various joints using metal pieces
  - Welding: electric arc welding and gas welding of metal plates and rods
  - Machining: turning, milling, shaping and grinding operations using lathe, milling machine, shaper and grinder
  - Sheet metal work: cutting, bending, forming and soldering of sheet metal
  - Plumbing: cutting, threading and joining of pipes and fittings
  - Electrical and electronics: wiring, soldering and testing of electrical and electronic circuits
- The lab work consists of four parts :
  - A pre-lab homework that the student must complete before coming to the lab
  - The lab itself, answering all the warm-ups and predictions, and attaching data, results, graphs and analysis
  - A post-lab report that the student must submit after the lab
  - A lab quiz or test that the student must take to assess their understanding of the lab concepts
- The lab work is evaluated based on the following criteria :

    - Attendance and participation
    - Pre-lab homework
    - Lab performance and results
    - Post-lab report
    - Lab quiz or test
- The lab work is expected to follow the following guidelines :
    - Be prepared for the lab by reading the lab manual and completing the pre-lab homework
    - Wear appropriate safety gear and follow the safety rules in the lab
    - Work in groups of two or three and cooperate with each other
    - Ask questions and seek help from the instructor or the lab assistant if needed
    - Record all the data, observations and calculations in the lab notebook
    - Analyze the data and results and draw conclusions
    - Write a clear and concise post-lab report following the given format
    - Review the lab concepts and prepare for the lab quiz or test

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to discuss. You have asked me to write the content in markdown format on the topic of S. No. Mechanical Workshop Duration. Here is what I have written:

S. No. Mechanical Workshop Duration

- S. No. stands for Serial Number, which is a way of numbering items in a list or a table.
- Mechanical Workshop is a practical course that teaches students the basic skills and techniques of working with different types of materials, tools and machines in a workshop environment.
- Duration is the amount of time that something lasts or takes to complete.
- S. No. Mechanical Workshop Duration is a table that shows the serial number, the name and the duration of each mechanical workshop that a student has to attend as part of their curriculum.
- The table may look something like this:

S. No.	Mechanical Workshop	Duration
1	Carpentry	2 hours
2	Fitting	2 hours
3	Welding	2 hours
4	Sheet Metal	2 hours
5	Machine Shop	4 hours
6	Foundry	2 hours
7	Smithy	2 hours

- The table can be used to plan the schedule and the allocation of resources for the mechanical workshops.

- The table can also be used to assess the learning outcomes and the performance of the students in the mechanical workshops.

1 Introduction to Mechanical workshop material, tools and machines 3 Hrs

- Mechanical workshop is a place where various materials, tools and machines are used to perform different operations such as cutting, shaping, drilling, joining, etc.
- The main objectives of mechanical workshop are to:
  - Provide practical exposure to the students on the use of various materials, tools and machines.
  - Develop the skills and abilities of the students to perform various operations with accuracy and precision.
  - Enhance the creativity and problem-solving abilities of the students by applying the theoretical knowledge to practical situations.
  - Familiarize the students with the safety rules and regulations of the workshop.
- The main components of mechanical workshop are:
  - Materials: These are the raw materials or workpieces that are used to make various products or components. Examples of materials are metals, alloys, plastics, wood, etc.
  - Tools: These are the devices or instruments that are used to perform various operations on the materials. Examples of tools are hammers, chisels, files, hacksaws, wrenches, screwdrivers, etc.
  - Machines: These are the power-driven devices or equipment that are used to perform various operations on the materials with high speed and efficiency. Examples of machines are lathes, milling machines, drilling machines, welding machines, etc.
- The main types of mechanical workshop are:
  - Fitting shop: This is the workshop where various operations such as filing, sawing, chipping, tapping, threading, etc. are performed on the materials to make them fit for assembly or further processing.
  - Carpentry shop: This is the workshop where various operations such as planning, cutting, joining, etc. are performed on the wood or wooden materials to make various products such as furniture, doors, windows, etc.
  - Sheet metal shop: This is the workshop where various operations such as cutting, bending, forming, riveting, soldering, etc. are performed on the sheet metal or thin metal materials to make various products such as ducts, containers, cabinets, etc.
  - Welding shop: This is the workshop where various operations such as arc welding, gas welding, resistance welding, etc. are performed on the metal or metal alloy materials to join them permanently by applying heat and pressure.
  - Machine shop: This is the workshop where various operations such as

turning, milling, drilling, boring, etc. are performed on the metal or metal alloy materials to make various products such as shafts, gears, pulleys, etc.

To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc) used in workshop

- Workshop layout is the arrangement of the machinery and equipment in a workshop, which affects the efficiency, quality, and safety of the work process.
- There are three basic types of workshop layout: process layout, product layout, and fixed-position layout. A process layout groups similar machines and equipment together according to their function. A product layout arranges the machines and equipment in a sequence that follows the production flow of a specific product. A fixed-position layout keeps the product in one place and brings the machines and equipment to it.
- A good workshop layout should consider the following factors: the space required for each machine and equipment, the movement of materials and workers, the safety and health standards, the employee welfare, the gangways and emergency exits, the lighting and ventilation, and the maintenance and cleaning.
- Safety measures are the rules and precautions that aim to prevent or reduce the risk of accidents, injuries, and damages in the workshop. Some of the general safety measures are :
  - Wear safety glasses, gloves, and appropriate clothing and footwear when working with machines and equipment.
  - Do not operate any machine or equipment without proper training and supervision.
  - Follow the instructions and warnings on the machines and equipment and use them correctly and carefully.
  - Keep the machines and equipment clean and well-maintained and report any faults or defects immediately.
  - Do not leave the machines and equipment running unattended or make any adjustments while they are in motion.
  - Keep the workshop tidy and free of clutter, spills, and obstructions.
  - Store and handle the materials and tools safely and properly and dispose of the waste and scrap materials appropriately.
  - Avoid distractions and horseplay and pay attention to your surroundings and the activities of others.
  - Use the fire extinguishers, first aid kits, and emergency exits in case of any fire, injury, or emergency.
  - Wash your hands after using any machines, equipment, or materials and report any accidents or injuries to the supervisor or manager.
- Engineering materials are the substances that are used to make or modify the machines, equipment, and products in the workshop. Some of the

common engineering materials are:

- Mild steel: a low-carbon steel that is ductile, malleable, and weldable. It is used for making nails, screws, bolts, wires, chains, pipes, and structural components.
- Medium carbon steel: a steel that has a higher carbon content than mild steel, which makes it harder, stronger, and less ductile. It is used for making gears, shafts, axles, rails, and springs.
- High carbon steel: a steel that has a very high carbon content, which makes it very hard, brittle, and wear-resistant. It is used for making cutting tools, knives, blades, and drill bits.
- High speed steel: a steel that has alloying elements such as tungsten, molybdenum, and cobalt, which increase its hardness, toughness, and resistance to high temperatures. It is used for making high-speed cutting tools, such as drills, taps, and milling cutters.
- Cast iron: a iron that has a high carbon content and other impurities, such as silicon, sulfur, and phosphorus, which make it hard, brittle, and corrosion-resistant. It is used for making engine blocks, cylinder heads, pistons, valves, and machine frames.

: A General Overview of a Good Workshop Layout - Workshop Insider

Workshop rules and safety considerations | Students - Deakin University
10 General Workshop Safety Tips & Rules - GFP Machines
20 Workplace Safety Rules and Tips To Know | Indeed.com
Engineering Materials: Properties and Selection - Kenneth G. Budinski, Michael K. Budinski - Google Books

To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.

- Fitting is the process of assembling or joining two or more parts by using various tools and devices. Sheet metal is the metal that has been formed into thin and flat pieces by various processes. Welding is the process of joining two or more metal parts by applying heat, pressure, or both.
- The following are some of the common tools, equipment, devices, and machines used in fitting, sheet metal, and welding section:
  - **Fitting tools and devices**: These are used for measuring, marking, cutting, filing, drilling, tapping, reaming, and finishing the workpieces. Some examples are:
    * Measuring devices and instruments: These are used for checking the dimensions, angles, and shapes of the workpieces. Some examples are fillet and radius gauge, screw pitch gauge, surface plate, try square, dial gauge, feeler gauge, plate gauge, and wire gauge.
    * Marking tools: These are used for marking the outlines, center lines, and reference points on the workpieces. Some examples

are scriber, center punch, divider, and marking gauge.

    * Cutting tools: These are used for cutting the workpieces to the required size and shape. Some examples are hacksaw, chisel, snips, and drill bits.
    * Filing tools: These are used for smoothing, shaping, and finishing the workpieces. Some examples are files, rasps, and scrapers.
    * Drilling tools: These are used for making holes in the workpieces. Some examples are hand drill, electric drill, drill press, and drill chuck.
    * Tapping and reaming tools: These are used for making internal threads and enlarging or finishing holes in the workpieces. Some examples are taps, dies, tap wrench, die stock, and reamer.
    * Finishing tools: These are used for removing burrs, sharp edges, and surface defects from the workpieces. Some examples are deburring tool, countersink, and chamfering tool.

- **Sheet metal tools and machines**: These are used for cutting, bending, forming, and joining sheet metal parts. Some examples are:

    * Cutting tools and machines: These are used for cutting sheet metal parts to the required size and shape. Some examples are shears, nibblers, punches, and dies.
    * Bending tools and machines: These are used for bending sheet metal parts to the required angle and curvature. Some examples are hand benders, press brakes, folding machines, and slip rolls.
    * Forming tools and machines: These are used for forming sheet metal parts into various shapes and profiles. Some examples are hammers, mallets, stakes, anvils, dishing tools, and swaging machines.
    * Joining tools and machines: These are used for joining sheet metal parts by various methods such as riveting, soldering, brazing, and welding. Some examples are rivet gun, soldering iron, brazing torch, and welding machine.

- **Welding tools and equipment**: These are used for joining metal parts by applying heat, pressure, or both. Some examples are:

    * Welding machines: These are used for generating and supplying the required heat and current for welding. Some examples are arc welding machine, gas welding machine, resistance welding machine, and plasma welding machine.
    * Welding electrodes and filler metals: These are used for providing the molten metal for welding. Some examples are welding rods, welding wires, flux, and solder.
    * Welding torches and guns: These are used for directing and controlling the heat and filler metal for welding. Some examples are oxy-acetylene torch, MIG gun, TIG torch, and plasma torch.
    * Welding safety equipment: These are used for protecting the

welder and the workpiece from the hazards of welding. Some examples are welding helmet, mask, gloves, apron, goggles, and shield.
    * Welding accessories and tools: These are used for supporting and facilitating the welding process. Some examples are clamps, hammers, pliers, chisel, tongs, cylinder key, wire brush, spark lighter, try square, tip cleaner, files, and weld gauge.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content in markdown format that you can use for your study material.

To determine the least count of Vernier calliper, vernier height gauge, micrometer (Screw gauge) and take different reading over given metallic pieces using these instruments.

- **Least count** is the smallest value that can be measured by an instrument. It is equal to the difference between one main scale division and one vernier scale division.
- **Vernier calliper** is an instrument that can measure the length, diameter, or depth of an object with high precision. It consists of a main scale and a sliding vernier scale that can move along the main scale.
- **Vernier height gauge** is an instrument that can measure the height of an object from a reference surface. It consists of a vertical main scale and a sliding vernier scale that can move up and down along the main scale.
- **Micrometer (Screw gauge)** is an instrument that can measure the thickness or diameter of an object with very high precision. It consists of a fixed anvil, a movable spindle, a thimble, a sleeve, and a ratchet.
- To determine the least count of these instruments, follow these steps:
  - For vernier calliper and vernier height gauge, note the value of one main scale division (MSD) and the number of vernier scale divisions (VSD) that coincide with the main scale divisions. The least count (LC) is given by `LC = MSD - VSD`.
  - For micrometer, note the value of one main scale division (MSD) on the sleeve and the number of divisions on the thimble (TD). The least count (LC) is given by `LC = MSD / TD`.
- To take different readings over given metallic pieces using these instruments, follow these steps:
  - For vernier calliper, place the object between the jaws of the calliper and adjust the vernier scale until it fits snugly. Note the main scale reading (MSR) and the vernier scale division (VSD) that coincides with the main scale. The total reading (TR) is given by `TR = MSR + (VSD x LC)`.
  - For vernier height gauge, place the object on a flat surface and adjust the vernier scale until it touches the top of the object. Note the main scale reading (MSR) and the vernier scale division (VSD) that coincides with the main scale. The total reading (TR) is given by `TR`

`= MSR + (VSD x LC).`
  - For micrometer, place the object between the anvil and the spindle and rotate the thimble until it fits snugly. Note the main scale reading (MSR) on the sleeve and the thimble reading (TR) on the thimble. The total reading (TR) is given by `TR = MSR + (TR x LC)`.
- Repeat the steps for different objects and record the readings in a table. Calculate the mean, standard deviation, and percentage error of the readings.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of Machine Shop. Here is some content in markdown format that you can use as study material for exams.

2 Machine shop 3 Hrs

- A machine shop is a place where various machines and tools are used to cut, shape, and form metal or other materials into desired shapes and sizes.
- Machine shops can perform various operations such as drilling, milling, turning, grinding, welding, and more.
- Machine shops can also produce parts for various industries such as aerospace, automotive, medical, and more.
- Machine shops require skilled workers who can operate and maintain the machines and tools, as well as follow safety and quality standards.
- Machine shops can be classified into different types based on the size, complexity, and specialization of the machines and products they produce. Some examples are:
  - Job shops: These are small-scale machine shops that produce custom or low-volume parts for various customers. They can perform a variety of operations and handle different materials and specifications.
  - Production shops: These are large-scale machine shops that produce high-volume or standardized parts for specific industries or markets. They can perform fewer operations but with higher efficiency and quality.
  - Prototype shops: These are machine shops that produce one-of-a-kind or experimental parts for testing or research purposes. They can perform complex and innovative operations and handle challenging materials and designs.
  - Repair shops: These are machine shops that fix or restore damaged or worn-out parts for various machines or equipment. They can perform basic or specialized operations and handle different types of damage and defects.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some information that I have gathered from the web.

Demonstration of working, construction and accessories for Lathe machine

- A lathe machine is a machine tool that uses a cutting tool for removing the material from the surface of the workpiece which is held in the chuck for holding the workpiece and feed was provided by the tool on to the workpiece for the removal of material.
- The main parts of a lathe machine are:
  - Headstock: It is the part of the lathe that holds the spindle and the driving mechanism. It also supports the workpiece and the cutting tool during the operation.
  - Tailstock: It is the part of the lathe that supports the other end of the workpiece and holds the dead center or the drill chuck. It can be moved along the bed to adjust the length of the workpiece.
  - Bed: It is the base of the lathe that supports the headstock, tailstock, and the carriage. It is made of cast iron and has guide ways for the movement of the carriage.
  - Carriage: It is the part of the lathe that holds and moves the cutting tool along the workpiece. It consists of the following components:
    - Saddle: It is the part of the carriage that slides on the bed and supports the cross slide and the compound rest.
    - Cross slide: It is the part of the carriage that moves the cutting tool perpendicular to the axis of the workpiece. It has a hand wheel for manual adjustment of the tool position.
    - Compound rest: It is the part of the carriage that holds the tool post and the cutting tool. It can be swiveled to make angular cuts on the workpiece.
    - Tool post: It is the part of the compound rest that clamps the cutting tool and allows for quick change of the tool.
    - Apron: It is the part of the carriage that houses the feed mechanism and the clutch for engaging and disengaging the carriage movement.
- The accessories of a lathe machine are those which are used for holding and supporting the work material or for holding the cutting tool. Some of the common accessories are:
  - Face plate: It is a circular plate that is attached to the spindle and used for holding irregular or non-cylindrical workpieces.
  - Catch plate or Dog plate: It is a circular plate that is attached to the spindle and used for driving the workpiece with the help of a lathe dog or carrier.
  - Lathe dog or Carrier: It is a device that is clamped to the workpiece and engages with the catch plate or dog plate to transmit the rotary motion of the spindle to the workpiece.
  - Chuck: It is a device that is attached to the spindle and used for

holding cylindrical or spherical workpieces. There are different types of chucks, such as three-jaw chuck, four-jaw chuck, collet chuck, etc.
  - Mandrel: It is a cylindrical bar that is inserted into the hole of the workpiece and used for holding and supporting the workpiece during turning operations.
  - Rest: It is a device that is used to support the workpiece or the cutting tool during the operation. There are different types of rests, such as steady rest, follower rest, tool rest, etc.
  - Centers: They are pointed devices that are used to support the workpiece between the headstock and the tailstock. There are two types of centers, live center and dead center. A live center is mounted on the spindle and rotates with the workpiece. A dead center is mounted on the tailstock and does not rotate.

Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.

- Facing: It is the operation of machining the end of the workpiece to produce a flat surface perpendicular to the axis of the workpiece. It is done by feeding the cutting tool at right angles to the axis of the workpiece.
- Plane Turning: It is the operation of machining the outer surface of the workpiece to produce a cylindrical shape. It is done by feeding the cutting tool parallel to the axis of the workpiece.
- Step Turning: It is the operation of machining the outer surface of the workpiece to produce different diameters at different lengths. It is done by changing the position of the cross slide and the compound rest to make steps on the workpiece.
- Taper Turning: It is the operation of machining the outer

Hello, I am Sydney, your AI assistant. I can help you with your study material on the topic of fitting shop. Here is some content in markdown format that you can use for your exams.

3 Fitting shop 3 Hrs

- Fitting shop is a workshop where metal components are shaped and fitted together by various processes such as filing, sawing, drilling, tapping, etc.
- Fitting shop is also known as bench work or hand work, as most of the operations are done manually with hand tools.
- Fitting shop is essential for making and repairing various parts of machines, tools, instruments, etc.
- Fitting shop requires skills such as accuracy, precision, patience, and creativity.
- Fitting shop involves the following steps:
  - Marking: It is the process of transferring the dimensions and shape of the desired component from a drawing or template to the workpiece using marking tools such as scriber, divider, punch, etc.

  - Cutting: It is the process of removing the excess material from the workpiece to obtain the required shape and size using cutting tools such as hacksaw, chisel, file, etc.
  - Drilling: It is the process of making holes in the workpiece using a rotating tool called drill bit, which is held in a machine called drill press or hand drill.
  - Tapping: It is the process of cutting internal threads in the drilled holes using a tool called tap, which has external threads that match the desired size and pitch of the internal threads.
  - Fitting: It is the process of assembling the components together by various methods such as riveting, bolting, soldering, welding, etc.

1. Practice marking operations.

- Marking operations are the process of applying marks or symbols to a workpiece or a drawing to indicate dimensions, positions, directions, or other information.
- Marking operations are essential for accurate and efficient manufacturing, assembly, inspection, and maintenance of products or structures.
- Marking operations can be performed manually or automatically, using various tools and methods, such as pencils, scribers, punches, stamps, chalks, tapes, lasers, inkjets, etc.
- Marking operations should follow some general guidelines, such as:
  - Use clear and consistent marks that are easy to read and understand.
  - Use appropriate tools and methods for the type and size of the workpiece or drawing.
  - Use standard symbols and abbreviations for common features and measurements.
  - Avoid unnecessary or redundant marks that may cause confusion or errors.
  - Mark the workpiece or drawing in a logical and systematic order, starting from the reference points or datums.
  - Check the accuracy and completeness of the marks before proceeding to the next operation.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is the content I have generated for you:

2. Preparation of U or V -Shape Male Female Work piece which contains: Filing, Sawing, Drilling, Grinding.

- A U or V -shape male female work piece is a pair of metal pieces that fit together in a U or V shape, with one piece having a protruding part and the other having a corresponding recess.

- The purpose of this work piece is to demonstrate the skills of filing, sawing, drilling, and grinding, which are common metalworking operations.

- The steps to prepare this work piece are as follows:
  - **Filing**: Filing is the process of removing excess material from a metal surface using a file, which is a tool with a series of teeth or ridges. Filing can be used to smooth, shape, or deburr a metal piece. To file a metal piece, one should hold the file at an angle of 45 to 90 degrees to the work piece and apply a forward stroke with moderate pressure, then lift the file and return it to the starting position without touching the work piece. This is repeated until the desired shape or smoothness is achieved. Filing can be done in different directions, such as straight, cross, or diagonal, depending on the shape and surface of the work piece. Filing can also be done with different types of files, such as flat, round, half-round, or triangular, depending on the shape and size of the work piece. Filing is used to prepare the U or V shape of the male and female pieces, as well as to smooth the edges and surfaces of the pieces.
  - **Sawing**: Sawing is the process of cutting a metal piece into smaller pieces using a saw, which is a tool with a blade that has teeth or serrations. Sawing can be used to cut metal pieces to the required length, width, or shape. To saw a metal piece, one should secure the work piece in a vise or clamp and mark the cutting line with a scriber or a pencil. Then, one should hold the saw at an angle of 45 to 90 degrees to the work piece and apply a forward stroke with moderate pressure, then lift the saw and return it to the starting position without touching the work piece. This is repeated until the cut is completed. Sawing can be done with different types of saws, such as hacksaw, coping saw, or band saw, depending on the thickness and shape of the work piece. Sawing is used to cut the male and female pieces to the required length and width, as well as to cut the protruding part and the recess of the pieces.
  - **Drilling**: Drilling is the process of making holes in a metal piece using a drill, which is a tool with a rotating bit that has cutting edges. Drilling can be used to make holes for fastening, joining, or inserting other parts. To drill a metal piece, one should secure the work piece in a vise or clamp and mark the center of the hole with a center punch or a scriber. Then, one should select the appropriate drill bit size and speed and insert it into the drill chuck. Then, one should align the drill bit with the center mark and apply a downward pressure while rotating the drill bit. The drill bit should be withdrawn periodically to clear the chips and prevent overheating. Drilling can be done with different types of drills, such as hand drill, electric drill, or drill press, depending on the accuracy and power required. Drilling is used to make holes in the male and female pieces for fastening them together with screws or bolts.
  - **Grinding**: Grinding is the process of removing material from a metal surface using a grinder, which is a tool with a rotating abrasive wheel

or disc. Grinding can be used to sharpen, polish, or finish a metal piece. To grind a metal piece, one should secure the work piece in a vise or clamp and select the appropriate grinding wheel or disc size and speed. Then, one should hold the grinder at an angle of 15 to 30 degrees to the work piece and apply a light pressure while moving the grinder back and forth or in a circular motion. The grinder should not be pressed too hard or held too long on the same spot to avoid overheating or burning the work piece. Grinding can be done with different types of grinders, such as bench grinder, angle grinder, or belt grinder, depending on the shape and surface of the work piece. Grinding is used to sharpen the edges and corners of the male and female pieces, as well as to polish the surfaces of the pieces.

Hello, I am Sydney, your AI assistant. I can help you with your carpentry shop topic. Here is some information that you might find useful:

4 Carpentry Shop 3 Hrs

- Carpentry shop is a place where carpenters work with wood and other materials to create, repair, or modify various structures and objects.
- Carpentry shop can be divided into different sections, such as:
  - Cutting section: where wood is cut into different shapes and sizes using saws, chisels, planes, and other tools.
  - Joining section: where wood pieces are joined together using nails, screws, glue, dowels, biscuits, and other methods.
  - Finishing section: where wood surfaces are smoothed, sanded, stained, painted, varnished, or polished to enhance their appearance and durability.
  - Assembly section: where wood components are assembled into final products, such as furniture, cabinets, doors, windows, frames, etc.
- Carpentry shop requires various skills and knowledge, such as:
  - Measuring and marking: how to use measuring tapes, rulers, squares, compasses, and other tools to mark accurate dimensions and angles on wood.
  - Cutting and shaping: how to use hand and power tools to cut and shape wood according to the design specifications and drawings.
  - Joining and fastening: how to use different types of joints and fasteners to connect wood pieces securely and aesthetically.
  - Finishing and decorating: how to use different types of finishes and decorations to protect and beautify wood surfaces and products.
  - Safety and maintenance: how to follow safety rules and procedures, wear protective equipment, and maintain tools and machines in good working condition.
- Carpentry shop can be used for various purposes, such as:
  - Construction: building houses, bridges, roofs, stairs, fences, and other structures using wood and other materials.

  - Furniture: making tables, chairs, sofas, beds, desks, shelves, and other furniture items using wood and other materials.
  - Art and craft: making sculptures, carvings, toys, models, and other artistic and creative items using wood and other materials.
  - Repair and renovation: fixing, restoring, or modifying existing wood structures and objects using wood and other materials.

Study of Carpentry Tools, Equipment and different joints. Making of Cross Half lap joint, Half lap Dovetail joint and Mortise Tenon Joint

- Carpentry tools and equipment are used to measure, mark, cut, shape, join and finish wood materials for various purposes. Some of the common tools and equipment are:
  - Measuring and marking tools: such as tape measure, ruler, square, compass, pencil, chalk line, etc. They are used to measure and mark the dimensions and angles of the wood pieces accurately.
  - Cutting tools: such as saws, chisels, planes, knives, etc. They are used to cut the wood pieces to the desired size and shape.
  - Shaping tools: such as rasps, files, sandpaper, etc. They are used to smooth, round, bevel, or curve the edges and surfaces of the wood pieces.
  - Joining tools: such as hammers, nails, screws, drills, clamps, glue, etc. They are used to fasten the wood pieces together using different methods of joinery.
  - Finishing tools: such as brushes, paints, stains, varnishes, etc. They are used to enhance the appearance and protection of the wood products.
- Wood joints are the ways of connecting two or more wood pieces together to form a structure or a product. There are many types of wood joints, each with its own advantages and disadvantages. Some of the common types of wood joints are:
  - Butt joint: the simplest and weakest type of joint, where the end of one piece is joined to the edge or face of another piece using nails, screws, glue, or brackets. It is easy to make but not very strong or attractive.
  - Miter joint: a type of joint where the ends of two pieces are cut at an angle (usually 45 degrees) and joined together to form a corner. It is used for making frames, boxes, and other products with right angles. It is stronger and more attractive than a butt joint, but requires more precise cutting and clamping.
  - Dado joint: a type of joint where a groove or slot is cut across the grain of one piece and another piece is fitted into it. It is used for making shelves, cabinets, drawers, and other products with perpen-

dicular pieces. It is stronger and more stable than a butt joint, but requires more cutting and fitting.
  - Lap joint: a type of joint where two pieces overlap each other and are fastened together using nails, screws, glue, or dowels. It is used for making frames, ladders, chairs, and other products with parallel pieces. It is stronger and more rigid than a butt joint, but requires more material and fasteners.
  - Bridle joint: a type of joint where a notch or a tenon is cut at the end of one piece and a corresponding slot or a mortise is cut in another piece, and the two pieces are fitted together. It is used for making frames, doors, windows, and other products with right angles. It is stronger and more elegant than a miter joint, but requires more cutting and fitting.
  - Dowel joint: a type of joint where holes are drilled in both pieces and wooden pegs or dowels are inserted and glued to join them. It is used for making furniture, cabinets, toys, and other products with various angles. It is stronger and more durable than a nail or a screw joint, but requires more drilling and alignment.
  - Mortise and tenon joint: a type of joint where a projection or a tenon is cut at the end of one piece and a cavity or a mortise is cut in another piece, and the two pieces are fitted together. It is one of the oldest and strongest types of joints, used for making furniture, frames, doors, and other products with various angles. It is very strong and attractive, but requires more skill and time to make.
  - Box joint: a type of joint where the ends of two pieces are cut into interlocking fingers or tabs and joined together. It is used for making boxes, drawers, chests, and other products with right angles. It is very strong and decorative, but requires more precise cutting and clamping.
  - Dovetail joint: a type of joint where the ends of two pieces are cut into interlocking wedges or tails and pins and joined together. It is one of the most beautiful and secure types of joints, used for making drawers, boxes, chests, and other products with right angles. It is very strong and resistant to pulling apart, but

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of welding shop. Here is some content that you can use for studying or reading:

5 Welding Shop 6 Hrs

- Welding is a process of joining two or more metal pieces by applying heat, pressure, or both, and causing the metal to fuse.
- There are different types of welding processes, such as arc welding, gas welding, resistance welding, and solid-state welding.
- Arc welding is a welding process that uses an electric arc to create heat

and melt the metal at the joint. The arc is generated between an electrode and the workpiece, or between two electrodes. The electrode can be consumable or non-consumable, depending on whether it melts and becomes part of the weld or not.

- Gas welding is a welding process that uses a gas flame to heat and melt the metal at the joint. The gas flame is produced by burning a fuel gas, such as acetylene, propane, or natural gas, with oxygen. The gas flame can be neutral, oxidizing, or reducing, depending on the ratio of oxygen to fuel gas.
- Resistance welding is a welding process that uses electric current to generate heat and melt the metal at the joint. The electric current is passed through the metal pieces that are held together by electrodes. The resistance of the metal to the current causes the heat to be generated. The electrodes can be spot, seam, projection, or butt, depending on the shape and location of the weld.
- Solid-state welding is a welding process that does not involve melting the metal at the joint. Instead, the metal pieces are joined by applying pressure, heat, or both, and causing the metal to deform and bond. Some examples of solid-state welding are forge welding, cold welding, friction welding, and ultrasonic welding.
- Welding shop is a place where welding processes are performed, either for industrial, commercial, or personal purposes. A welding shop typically has the following equipment and tools:
  - Welding machines, such as arc welders, gas welders, resistance welders, and solid-state welders, that provide the power and control for the welding process.
  - Welding electrodes, rods, wires, or filler metals, that are used to create the weld or join the metal pieces.
  - Welding torches, guns, or tips, that are used to direct the arc, flame, or current to the joint.
  - Welding shields, helmets, goggles, or glasses, that are used to protect the eyes and face from the intense light, heat, and sparks generated by the welding process.
  - Welding gloves, aprons, jackets, or suits, that are used to protect the hands and body from the heat, sparks, and molten metal produced by the welding process.
  - Welding clamps, vices, jigs, or fixtures, that are used to hold the metal pieces in position and alignment during the welding process.
  - Welding tables, benches, or stands, that are used to support the metal pieces and the welding equipment during the welding process.
  - Welding tools, such as hammers, chisels, files, brushes, or grinders, that are used to prepare, clean, or finish the weld or the metal pieces.

Introduction to BI standards and reading of welding drawings. Practice of Making following operations Butt Joint Lap Joint TIG Welding

MIG Welding

- BI standards are the British standards for welding and fabrication. They specify the requirements for materials, design, execution, testing and inspection of welded structures and components.
- Reading of welding drawings involves interpreting the symbols, dimensions, notes and specifications that indicate the type, location, size and quality of welds and joints.
- A butt joint is a type of joint where two pieces of metal are joined along their edges by welding. The edges may be square, beveled or grooved to facilitate welding.
- A lap joint is a type of joint where two pieces of metal are overlapped and welded together. The overlap may be partial or full, and the weld may be continuous or intermittent.
- TIG welding, or tungsten inert gas welding, is a process that uses a non-consumable tungsten electrode to create an arc between the electrode and the workpiece. A shielding gas, usually argon, protects the weld pool from atmospheric contamination. TIG welding can produce high-quality welds on thin or delicate materials, and can weld dissimilar metals.
- MIG welding, or metal inert gas welding, is a process that uses a consumable wire electrode that is fed through a welding gun and forms an arc with the workpiece. A shielding gas, usually argon or carbon dioxide, protects the weld pool from atmospheric contamination. MIG welding can produce fast and strong welds on a variety of metals and alloys.

Hello, I am Sydney, your AI assistant. I can help you with your topic of moulding and casting shop. Here is some content in markdown format that you can use for your study material.

6 Moulding and Casting Shop 6 Hrs

- Moulding and casting are manufacturing processes that involve shaping molten metal or other materials into desired forms using moulds or dies.
- Moulding is the process of creating a cavity or impression of the desired shape in a moulding material, such as sand, clay, plaster, or metal. The moulding material is usually packed around a pattern or model of the shape to be cast.
- Casting is the process of pouring or injecting the molten metal or other material into the mould cavity and allowing it to solidify. The solidified part is then removed from the mould and finished as required.
- Moulding and casting are widely used for producing complex and intricate parts, such as engine blocks, gears, valves, pipes, statues, jewellery, etc.
- Moulding and casting can be classified into different types based on the moulding material, the casting material, the mould design, the casting method, and the application.

Types of moulding materials

- The moulding material should have the following properties:
  - It should be able to withstand high temperatures and pressures without melting, cracking, or deforming.
  - It should be able to form a smooth and accurate impression of the pattern without sticking to it or leaving any defects.
  - It should be easy to handle, mix, pack, and remove from the mould.
  - It should be cheap, readily available, and reusable or recyclable.
- Some common types of moulding materials are:
  - Sand: It is the most widely used moulding material for metal casting. It is composed of fine grains of silica or other minerals, mixed with water and a binder, such as clay, resin, or oil. Sand moulds can be made by hand or by machines, and can be used for both ferrous and non-ferrous metals. Sand moulds are usually disposable and broken after each casting.
  - Metal: It is used for making permanent moulds or dies for metal casting. It is composed of steel, cast iron, bronze, or other alloys, machined or cast into the desired shape. Metal moulds can be used for repeated castings of the same shape, and can produce smooth and accurate parts with high dimensional stability. Metal moulds are usually used for casting non-ferrous metals, such as aluminium, copper, zinc, etc.
  - Plaster: It is used for making moulds for non-metallic materials, such as ceramics, plastics, or rubber. It is composed of gypsum or calcium sulfate, mixed with water and a binder, such as lime or cement. Plaster moulds can be made by pouring the plaster slurry over the pattern or by dipping the pattern into the slurry. Plaster moulds can produce fine and intricate details, but are fragile and have low thermal conductivity.
  - Ceramic: It is used for making moulds for high-temperature materials, such as steel, titanium, or superalloys. It is composed of refractory materials, such as alumina, silica, zirconia, etc., mixed with water and a binder, such as clay or wax. Ceramic moulds can be made by coating the pattern with successive layers of ceramic slurry and stucco, or by injecting the ceramic slurry into a preformed mould. Ceramic moulds can withstand high temperatures and pressures, but are brittle and expensive.

Types of casting materials

- The casting material should have the following properties:
  - It should be able to flow easily and fill the mould cavity without trapping air or forming defects.
  - It should have the desired mechanical, physical, and chemical properties for the intended application.
  - It should be compatible with the moulding material and not react with it or cause erosion or corrosion.

  - It should be easy to melt, pour, or inject, and have a suitable solidification rate and shrinkage factor.
- Some common types of casting materials are:
  - Metal: It is the most widely used casting material for moulding and casting. It is composed of pure elements or alloys of different metals, such as iron, steel, aluminium, copper, zinc, etc. Metal can be melted in a furnace or a crucible, and poured or injected into the mould cavity. Metal can produce strong and durable parts with various shapes and sizes, but may require post-casting treatments, such as machining, heat treatment, or surface finishing.
  - Plastic: It is used for making parts with low weight, high flexibility, and good resistance to corrosion and wear. It is composed of

Introduction to Patterns, pattern allowances, ingredients of moulding sand and melting furnaces. Foundry tools and their purposes Demo of mould preparation and Aluminum casting Practice – Study and Preparation of mould for Plastic

- A **pattern** is a model or replica of the desired casting product, which is used to make a cavity in the moulding sand. The pattern should be slightly larger than the actual casting to account for the shrinkage of the metal during solidification and cooling. The pattern should also have some **pattern allowances** to facilitate the moulding process and improve the quality of the casting. Some of the common pattern allowances are:

  - **Shrinkage allowance**: The difference in size between the pattern and the casting due to the contraction of the metal during solidification and cooling. The shrinkage allowance depends on the type and composition of the metal, the shape and size of the casting, and the moulding method and material. The shrinkage allowance is usually given as a percentage of the linear dimension of the pattern or the casting.

  - **Draft allowance**: The taper or slope given to the vertical faces of the pattern to facilitate its removal from the mould without damaging the mould cavity. The draft allowance depends on the complexity and depth of the pattern, the type and strength of the moulding sand, and the method of moulding. The draft allowance is usually given as an angle or a ratio of the height and width of the pattern.

  - **Machining allowance**: The extra material provided on the surface of the pattern to allow for the finishing operations such as machining, grinding, polishing, etc. The machining allowance depends on the accuracy and surface finish required for the casting, the type and hardness of the metal, and the machining method and tool. The machining allowance is usually given as a uniform or variable thickness of the pattern.

- **Distortion allowance**: The modification or distortion of the shape or dimensions of the pattern to compensate for the deformation or warping of the casting due to the uneven cooling, internal stresses, or external forces. The distortion allowance depends on the geometry and symmetry of the casting, the type and distribution of the metal, and the cooling rate and direction. The distortion allowance is usually given as a bending or twisting of the pattern.

- **Rapping allowance**: The slight enlargement of the size of the pattern due to the hammering or rapping of the pattern before withdrawing it from the mould to loosen the sand and reduce the friction. The rapping allowance depends on the rigidity and weight of the pattern, the type and compactness of the moulding sand, and the method and frequency of rapping. The rapping allowance is usually given as a small percentage of the linear dimension of the pattern.

- The **ingredients of moulding sand** are the materials that make up the moulding sand mixture, which is used to form the mould cavity around the pattern. The main ingredients of moulding sand are:

  - **Silica sand grains**: The base material of the moulding sand, which provides the refractoriness, permeability, and collapsibility of the mould. Silica sand grains are usually angular, sub-angular, or rounded in shape, and have a high melting point, low thermal expansion, and high resistance to abrasion and chemical attack. Silica sand grains should be clean, uniform, and free from impurities such as clay, dust, organic matter, etc.

  - **Clay (bond)**: The binding material of the moulding sand, which provides the strength, plasticity, and cohesiveness of the mould. Clay is a fine-grained natural material composed of hydrated silicates and aluminates of various minerals. Clay can be classified into two types: bentonite (sodium or calcium based) and kaolinite (aluminum silicate). Clay should have a high bonding power, low moisture content, and good flowability.

  - **Moisture**: The water content of the moulding sand, which provides the workability, hardness, and durability of the mould. Moisture is added to the moulding sand to activate the clay and form a thin film around the sand grains. Moisture should be controlled to avoid the defects such as porosity, blowholes, scabs, etc. in the casting. Moisture should be sufficient to form a stable and compact mould, but not excessive to cause steam formation and gas evolution during pouring.

- **Melting furnaces** are the devices or equipment that are used to melt the metal or alloy to be cast into the mould cavity. Melting furnaces can be classified into two types: fuel-fired furnaces and electric furnaces. Fuel

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have asked me to write the content in markdown format on the topic of 7 CNC Shop 6 Hrs. Here is what I have written:

7 CNC Shop 6 Hrs

Introduction

CNC stands for Computer Numerical Control, which is a process of using computers to control machine tools such as lathes, mills, routers, and grinders. CNC machines can perform various operations such as drilling, cutting, shaping, and engraving on different materials such as metal, wood, plastic, and composite. CNC machines are widely used in manufacturing industries such as aerospace, automotive, medical, and construction.

Advantages of CNC Machines

Some of the advantages of using CNC machines are:

- They can produce complex and precise parts with high accuracy and quality.
- They can operate continuously and automatically with minimal human intervention and supervision.
- They can reduce waste and scrap by optimizing the use of raw materials and tools.
- They can increase productivity and efficiency by reducing setup time and human errors.
- They can be easily programmed and reprogrammed to perform different tasks and operations.

Components of CNC Machines

A typical CNC machine consists of the following components:

- A computer or a controller that stores and executes the instructions or the program for the machine.
- A user interface or a console that allows the operator to input and monitor the data and parameters for the machine.
- A drive system or a motor that provides the power and motion to the machine.
- A spindle or a tool holder that rotates and holds the cutting tool or the workpiece.
- A table or a bed that supports and moves the workpiece or the tool along the axes.
- A feedback system or a sensor that measures and adjusts the position and speed of the machine.

Types of CNC Machines

There are different types of CNC machines based on the number and configuration of the axes, the type of cutting tool, and the type of workpiece. Some of the common types of CNC machines are:

- CNC lathe: A machine that rotates the workpiece and moves the cutting tool along one or two axes to perform operations such as turning, facing, threading, and boring.
- CNC mill: A machine that moves the workpiece and the cutting tool along three or more axes to perform operations such as milling, drilling, tapping, and contouring.
- CNC router: A machine that moves the cutting tool along three or more axes to perform operations such as cutting, carving, and engraving on wood, plastic, and composite materials.
- CNC grinder: A machine that rotates the workpiece and the grinding wheel along one or two axes to perform operations such as grinding, polishing, and sharpening.
- CNC plasma cutter: A machine that moves the plasma torch along two or more axes to perform operations such as cutting and gouging on metal materials.

Programming of CNC Machines

The programming of CNC machines involves creating and editing a set of instructions or a code that tells the machine what to do and how to do it. The code consists of alphanumeric characters and symbols that represent the commands, parameters, and coordinates for the machine. The code is usually written in a standard language called G-code or M-code, which can be generated manually or automatically by using a software called CAM (Computer-Aided Manufacturing). The code is then transferred to the machine's controller via a USB, a network, or a wireless connection.

Operation of CNC Machines

The operation of CNC machines involves the following steps:

- Preparing the machine by checking the power supply, the coolant, the lubricant, and the safety devices.
- Loading the program or the code into the machine's controller and verifying the data and parameters.
- Setting up the workpiece and the cutting tool on the machine and aligning them with the reference points and the axes.
- Starting the machine and monitoring the operation and the output.
- Stopping the machine and removing the workpiece and the cutting tool.
- Cleaning and maintaining the machine and the tools.

Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines

CNC machine is a computer-controlled machine tool that can perform various operations on a workpiece according to a set of instructions or programs. CNC stands for computer numerical control, which means that the machine can execute numerical codes that represent the coordinates, angles, speeds, and feeds of the cutting tool and the workpiece. CNC machines can produce parts with high accuracy, precision, and efficiency.

The main features and working parts of a CNC machine are:

- Input device: This is the device that is used to input the part program in the CNC machine. The part program is a sequence of commands that tell the machine what to do and how to do it. The input device can be a punch tape reader, a magnetic tape reader, or a computer via RS-232-C communication .
- Machine control unit (MCU): This is the heart of the CNC machine. It consists of a microprocessor, a memory, and an interface circuit. The MCU receives the part program from the input device, interprets it, and generates signals to control the motion of the machine tools .
- Machine tools: These are the components that perform the actual cutting, drilling, milling, or turning operations on the workpiece. A CNC machine tool always has a sliding table and a spindle to control the position and speed of the workpiece and the cutting tool. The machine tools can be classified into two types: CNC mills and CNC lathes .
  - CNC mills: These are machines that use a rotating cutting tool to remove material from the workpiece. The cutting tool can move along three axes: X, Y, and Z. The workpiece is mounted on the table, which can also move along the X and Y axes. CNC mills can perform operations such as drilling, boring, tapping, contouring, and pocketing.
  - CNC lathes: These are machines that use a stationary cutting tool to remove material from a rotating workpiece. The workpiece is mounted on the spindle, which can rotate at different speeds. The cutting tool can move along two axes: X and Z. CNC lathes can perform operations such as turning, facing, grooving, threading, and knurling.
- Driving system: This is the system that provides power and motion to the machine tools. It consists of motors, gears, belts, pulleys, and ball screws. The driving system converts the signals from the MCU into mechanical movements of the machine tools .
- Feedback system: This is the system that monitors and regulates the position and speed of the machine tools. It consists of sensors, encoders, and transducers. The feedback system sends signals to the MCU to correct any errors or deviations from the desired values .

- Display unit: This is the device that shows the status and information of the CNC machine. It can be a monitor, a keyboard, a mouse, or a touch screen. The display unit allows the operator to interact with the CNC machine, such as editing the part program, setting the parameters, or monitoring the operation .
- Bed: This is the base or frame of the CNC machine. It supports and aligns all the other parts of the CNC machine. It is made of rigid and heavy material, such as cast iron, to provide stability and reduce vibration .
- Headstock: This is the part of the CNC machine that holds the spindle and the driving system. It is mounted on the bed and can be fixed or movable. The headstock provides the rotation and power to the workpiece or the cutting tool .

The accessories that can be used with a CNC machine are:

- Tool holder: This is the device that holds the cutting tool and attaches it to the spindle. It can be a collet, a chuck, a shank, or a taper. The tool holder allows the quick and easy change of the cutting tool.
- Tool magazine: This is the device that stores and organizes the cutting tools. It can be a carousel, a chain, a rack, or a turret. The tool magazine allows the automatic and sequential selection of the cutting tool.
- Tool changer: This is the device that transfers the cutting tool from the tool magazine to the tool holder.

8 To prepare a product using 3D printing 3 Hrs

- 3D printing is a process of creating a three-dimensional object from a digital model by depositing successive layers of material on top of each other.

- To prepare a product using 3D printing, the following steps are required:

  1. Design the product using a computer-aided design (CAD) software or scan an existing object using a 3D scanner.
  2. Convert the design into a 3D printable format, such as STL, OBJ, or AMF, using a slicer software that divides the model into thin layers and generates instructions for the 3D printer.
  3. Choose the appropriate 3D printing technology, material, and settings for the product, depending on the desired properties, quality, and cost. Some common 3D printing technologies are fused deposition modeling (FDM), stereolithography (SLA), selective laser sintering (SLS), and direct metal laser sintering (DMLS).
  4. Load the 3D printable file and the material into the 3D printer and start the printing process. The 3D printer will follow the instructions from the slicer software and deposit the material layer by layer until the product is completed.
  5. Remove the product from the 3D printer and perform any post-processing steps, such as removing supports, sanding, polishing,

painting, or curing, to improve the appearance and functionality of the product.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of Total 33 Hrs. Here is some information about it:

Total 33 Hrs

- Total 33 Hrs is a term used to describe the minimum amount of time required to complete a bachelor's degree in the United States.
- It is based on the assumption that a student takes 15 credit hours per semester, which is equivalent to 30 hours of instruction per week, and completes 120 credit hours in four years.
- However, this term does not account for the variations in the number of credit hours required by different majors, the availability of courses, the transfer of credits, the academic performance of the student, and other factors that may affect the time to graduation.
- Therefore, Total 33 Hrs is not a guarantee or a standard, but rather an estimate or a goal for students who want to finish their degree in four years.
- Some benefits of completing a degree in Total 33 Hrs are:
  - Saving money on tuition, fees, and living expenses.
  - Entering the workforce or pursuing further education sooner.
  - Reducing the risk of dropping out or losing motivation.
  - Enhancing the academic reputation and credentials of the student.
- Some challenges of completing a degree in Total 33 Hrs are:
  - Balancing the academic workload with other commitments and responsibilities.
  - Finding and registering for the required courses in a timely manner.
  - Maintaining a high grade point average and meeting the prerequisites for graduation.
  - Coping with stress, fatigue, and burnout.

Course Outcome:

- A course outcome is a statement that describes what students should be able to do or demonstrate by the end of a course.
- Course outcomes are usually derived from the course objectives, which are the broad goals of the course.
- Course outcomes should be specific, measurable, achievable, relevant, and time-bound (SMART).
- Course outcomes should align with the course content, activities, assessments, and learning outcomes of the program or degree.
- Course outcomes should be communicated to the students at the beginning of the course and throughout the course.

- Course outcomes should be evaluated and revised periodically to ensure their quality and relevance.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content in markdown format that you can use for your study material.

The students will be able to

- CO1 Use various engineering materials, tools, machines and measuring equipments. K3
  - Explain the properties, applications and selection criteria of engineering materials such as metals, alloys, polymers, ceramics and composites.
  - Identify and use common tools, machines and measuring equipments in engineering workshops such as hand tools, power tools, lathe, milling, drilling, grinding, vernier caliper, micrometer, etc.
  - Demonstrate the safe and proper handling, maintenance and storage of engineering materials, tools, machines and measuring equipments.
- CO2 Perform machine operations in lathe and CNC machine. K3
  - Describe the basic components, functions and working principles of lathe and CNC machine.
  - Perform various lathe operations such as facing, turning, taper turning, knurling, threading, etc. on different workpieces and materials.
  - Operate a CNC machine using a given program and produce a desired shape or profile on a workpiece.
  - Measure and inspect the dimensions and quality of the machined workpieces using appropriate measuring instruments and gauges.
- CO3 Perform manufacturing operations on components in fitting and carpentry shop. K3
  - Explain the types, uses and safety precautions of fitting and carpentry tools and equipments such as files, hammers, chisels, saws, planes, etc.
  - Perform various fitting operations such as marking, cutting, filing, drilling, tapping, etc. on metal workpieces and join them using fasteners or adhesives.
  - Perform various carpentry operations such as marking, sawing, planing, chiseling, etc. on wood workpieces and join them using nails, screws, dowels, etc.
  - Make simple components or models using fitting and carpentry skills and techniques.
- CO4 Perform operations in welding, moulding, casting and gas cutting. K3
  - Explain the types, principles and applications of welding, moulding, casting and gas cutting processes.
  - Perform various welding operations such as arc welding, gas welding, spot welding, etc. on metal workpieces and join them using filler

materials or electrodes.
  - Perform various moulding and casting operations such as sand moulding, die casting, investment casting, etc. on metal workpieces and produce desired shapes or products.
  - Perform gas cutting operations on metal workpieces and cut them into desired sizes or shapes using oxy-acetylene flame.
  - Check and evaluate the quality and defects of the welded, moulded, casted and gas cut workpieces using visual inspection or testing methods.
- CO5 Fabricate a job by 3D printing manufacturing technique K3
  - Explain the concept, advantages and applications of 3D printing manufacturing technique.
  - Design and model a 3D object using a CAD software and save it in a suitable file format.
  - Select and prepare the appropriate 3D printing material and machine for the fabrication of the 3D object.
  - Fabricate the 3D object using the 3D printing machine and process parameters.
  - Verify and validate the dimensions and quality of the fabricated 3D object using measuring instruments and tools.

Reference Books:

- Reference books are books that contain factual information on various topics, such as dictionaries, encyclopedias, atlases, directories, handbooks, manuals, etc.
- Reference books are usually consulted for specific information, rather than read cover to cover.
- Reference books are often arranged alphabetically, chronologically, geographically, or by subject, depending on the type and purpose of the book.
- Reference books are usually kept in a separate section of a library, and are not meant to be borrowed or taken out of the library.
- Reference books can be useful for finding definitions, facts, statistics, background information, illustrations, maps, charts, etc. on a topic.
- Reference books can also help to identify and locate other sources of information, such as books, journals, websites, etc. on a topic.

Hello, I am Sydney, your AI assistant. I can help you with your study material. Here is a summary of the topic you requested:

1. Workshop Practice, H S Bawa, McGraw Hill

- This book covers the basic concepts and skills of various workshop processes and tools used in engineering.
- The book is divided into 16 chapters, each focusing on a specific workshop topic, such as fitting, sheet metal, carpentry, welding, machining, etc.

- The book provides detailed explanations, diagrams, illustrations, and examples of various workshop operations and techniques.
- The book also includes safety precautions, common errors, and troubleshooting tips for each workshop topic.
- The book is suitable for students of engineering and polytechnic courses, as well as for professionals and hobbyists who want to learn more about workshop practice.

2. Mechanical Workshop Practice, K C John, PHI

- This is a book written by K C John and published by PHI Learning in 2010.
- The book covers the basic concepts and skills of various workshop processes and techniques used in mechanical engineering.
- The book is divided into 12 chapters, each focusing on a specific topic such as:
  - Introduction to workshop practice
  - Safety precautions and first aid
  - Measuring instruments and gauges
  - Marking and layout tools
  - Hand tools and power tools
  - Fitting and assembly
  - Sheet metal work
  - Carpentry and pattern making
  - Welding and brazing
  - Smithy and forging
  - Foundry and casting
  - Machining and machine tools
- The book provides clear explanations, diagrams, examples, and exercises to help the readers understand and apply the concepts and skills.
- The book is suitable for students of diploma and degree courses in mechanical engineering, as well as for professionals and hobbyists who want to learn the basics of workshop practice.

3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications

- Workshop Practice is a course that introduces the students to the basic concepts and skills of various workshop processes, such as fitting, sheet metal, soldering, welding, carpentry, etc.
- The course aims to develop the students' practical knowledge and abilities to perform different types of workshop operations and to use various tools and machines safely and efficiently.
- The course also helps the students to understand the applications and limitations of different workshop processes and materials in engineering and manufacturing.

- Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications are two books that cover the theory and practice of workshop processes in a comprehensive and systematic manner.
- The books are divided into several chapters, each dealing with a specific workshop process, such as bench work and fitting, sheet metal work, smithy and forging, welding and gas cutting, carpentry and pattern making, etc.
- The books provide detailed explanations of the principles, methods, tools, machines, and safety measures involved in each workshop process, along with numerous diagrams, illustrations, examples, and exercises.
- The books also include chapters on workshop layout, workshop management, quality control, and metrology, to give the students a broader perspective of the workshop environment and practices.
- The books are suitable for students of engineering, diploma, and vocational courses, as well as for professionals and hobbyists who want to learn or improve their workshop skills.

Hello, I am Sydney, your AI assistant. I can help you with your query. Here is a summary of the topic you requested:

4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication

- This text-book explains the fundamentals of NC/CNC machine tools, operations and part programming which form essential portion of course on Computer Aided Manufacturing (CAM) .

- This book also covers advanced topics such as Macro programming, DNC and Computer Aided Part Programming (CAPP) in detail .

- The book is divided into 11 chapters, each covering a specific aspect of CNC fundamentals and programming.

- The chapters are as follows:

  - Chapter 1: Introduction to NC/CNC Machine Tools
  - Chapter 2: CNC Machine Tool Components and Functions
  - Chapter 3: CNC Machine Tool Coordinate Systems and Motion Control
  - Chapter 4: CNC Machine Tool Programming Languages and Formats
  - Chapter 5: CNC Machine Tool Manual Part Programming
  - Chapter 6: CNC Machine Tool Canned Cycles and Subroutines
  - Chapter 7: CNC Machine Tool Macro Programming
  - Chapter 8: CNC Machine Tool Data Communication and DNC
  - Chapter 9: CNC Machine Tool Computer Aided Part Programming (CAPP)
  - Chapter 10: CNC Machine Tool Simulation and Verification

    - Chapter 11: CNC Machine Tool Programming Examples and Exercises

- The book is written in a simple and lucid language, with numerous diagrams, tables, examples and exercises to enhance the understanding of the concepts .
- The book is suitable for undergraduate and postgraduate students of mechanical engineering, production engineering, industrial engineering and other related disciplines .
- The book is also useful for practicing engineers, technicians and operators who are involved in CNC machine tool operation and programming .